

Vipin Kumar

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SUMMARY

Embedded Systems Software Engineer with 2+ years at KPIT Technologies delivering production automotive infotainment (IVI) features in modern C++ across two OEM platforms. Owns end-to-end feature delivery — from Bluetooth connectivity stacks (HFP, PBAP, MAP, BLE) to AOSP-based camera UI features — including production-shipped code, systematic defect resolution, and test infrastructure. Converted from trainee to full-time and promoted to Engineer within three years at the same company. Active open-source contributor to C++ testing and systems-infrastructure tools.

WORK EXPERIENCE

KPIT Technologies — Pune, India

Engineer (Feb 2026 – Present) · Associate Engineer (Feb 2025 – Jan 2026) · Trainee (Dec 2023 – Jan 2025) — converted to full-time (Feb 2025), promoted to Engineer (Feb 2026)

Telechip AOSP Infotainment Project | Jun 2026 – Present

- Independently designed and implemented the reverse-camera parking-guideline overlay feature at the application layer on a TCC805x-based AOSP infotainment platform; shipped to production for an OEM client.
- Solely triaged and resolved ~20 Bluetooth defects spanning connection drops, pairing failures, audio routing, and HFP/PBAP issues, working within a 15-engineer cross-functional team in direct collaboration with the OEM client.

Bluetooth Connectivity Project | Feb 2025 – Jun 2026

- Designed and developed reusable C++ modules for core Bluetooth connectivity profiles (HFP, PBAP, MAP, BLE) within an automotive infotainment head unit, collaborating directly with OEM clients to ensure seamless feature delivery.
- Performed systematic root cause analysis and resolved critical software defects through comprehensive log analysis, including HCI and Bluetooth sniffer logs.
- Authored and maintained GTest unit tests covering 10+ multithreaded and embedded system components, improving software reliability and test coverage.
- Automated the Linux-based build, deployment, and file-transfer pipeline for the Phone Interactor (PHNI) binary using Docker and shell scripting, collapsing a 5-step manual process into a single-command workflow.
- Applied modern C++14/17 features including RAII, move semantics, and concurrency primitives within production codebases while collaborating across cross-functional Agile/Scrum teams.

Trainee | Dec 2023 – Jan 2025

- Ramped up on modern C++ (C++14/17), multithreaded/embedded systems, socket programming, and Agile/Scrum and Git-based workflows; converted to full-time Associate Engineer based on performance.

PROJECTS

Inventory Scanner — Android App | Android (Java), Apache POI, QR APIs | Internal Tool

- Built and deployed an Android application across multiple ODC teams, replacing a manual 2–3 hour hardware audit process with an automated QR-based workflow that reduced audit time by over 80%.

OPEN SOURCE CONTRIBUTIONS

EventLoop — C++17, GTest, CMake, Icov, OpenCppCoverage | 2 PRs

- PR #4: Added GTest unit tests from scratch covering 6 core EventLoop APIs; achieved 67.5% line / 84.8% function coverage with CMake restructuring and Windows/MSVC cross-platform support.
- PR #5: Improved test coverage to 88.6% lines / 100% functions; automated Windows coverage report generation via OpenCppCoverage integrated with a cross-platform build.sh pipeline.

Bitmask — Python, OpenSSL, PKI

- Contributed Windows portability support to a hybrid-encryption (RSA-2048 + AES-256) secure file-sharing tool, extending compatibility beyond the existing Linux/Mac support baseline.

SKILLS SUMMARY

- Programming Languages:** C++, C, Java, Python
- C++ & Systems Programming:** Multithreading, Memory Management, RAII, Move Semantics, std::atomic, Lock-free Programming, STL, Socket Programming
- Bluetooth, Embedded, AOSP & Linux:** IVI Systems, Bluetooth Classic & BLE, HFP, PBAP, MAP, PHNI, Bluetooth Sniffer Logs, HCI Log Analysis, AOSP (application layer), TCC805x, Embedded C++, Linux (Ubuntu), IPC, ADB
- Testing, Tools & Processes:** GTest, CMake, Git, GitHub Actions, Docker, Shell Scripting, OpenCppCoverage, Icov, Static Analysis, Root Cause Analysis, Software Debugging, Agile/Scrum, Software Specification Review

EDUCATION & ACHIEVEMENTS

Bachelor of Technology, Computer Science (Specialization: Artificial Intelligence & Machine Learning) | Chandigarh Engineering College, Chandigarh, India | June 2020 – August 2024

- Secured 2nd place in the C++ track at Code Vista, KPIT's internal coding competition.